

Project Performance Audit & Cost Report

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Executive Summary

This is a Residential building with 5 structural divisions, including 2 bedrooms, 2 bathrooms, 1 kitchen, and 1 hall. Based on the uploaded blueprint, the estimated construction cost is ₹79,97,479 using Standard quality specifications. The estimated completion time is 7 months.

Project Details & Assigned Personnel

Project Name:	Apex Test Residency 2	Client Name:	Apex Builders
Location:	Mumbai, Maharashtra	Building Type:	Residential
Floors:	2	Material Quality:	Standard
Assigned PM:	Marcus Aurelius	Assigned Engineer:	Diana Prince

AI Blueprint Layout Analysis

Structural Division	Detected Quantity	Classification
Bedrooms / Cabin Areas	2 Zones	Enclosure Space
Bathrooms / Plumbing	2 Zones	Sanitary Block
Kitchen / Cafeteria	1 Zone	Services Block
Hall / Reception Zone	1 Zone	Common Area
Dining / Pantry Area	1 Zone	Amenity space
Balconies	2 Zones	Open Air Extension
Parking Slots	1 Zone	Utility
Staircase / Lift Core	1 Zone	Vertical Circulation
Doors Identified	8 Units	Apertures / Access
Windows Identified	12 Units	Apertures / Ventilation
Total Built-up Area	2,400 sqft	Calculated Area

AI Material Quantity & Sourcing Schedule

Material Item	Estimated Qty	Unit	Total Cost (INR)
Cement	960	bags	₹3,64,800
Steel Rebar	9.6	tons	₹5,95,200
Bricks	52,800	pcs	₹4,22,400
Sand	192	tons	₹2,30,400
Blue Metal Aggregate	144	tons	₹2,01,600
Gravel / Soling	96	tons	₹96,000
Ready-Mix Concrete	72	cum	₹3,24,000
Paint	384	liters	₹99,840
Flooring Tiles	2,760	sqft	₹1,79,400
Electrical Conduit & Wiring	1	lump-sum	₹2,16,000
Plumbing Pipes & Fixtures	1	lump-sum	₹1,92,000
Roofing/Slab Concrete reinforcement	1	lump-sum	₹3,12,000
Doors	8	units	₹48,000
Windows	12	units	₹48,000
Labor Sourcing Requirement	2,040	man-days	₹11,22,000

Financial Summary & Cost Breakdown

Cost Category	Estimated Cost (INR)	Cost Category	Estimated Cost (INR)
Materials Base Cost	₹33,29,640	Labor Cost	₹11,22,000
Equipment & Machinery	₹4,80,000	Electrical Works	₹2,16,000
Plumbing & Sewerage	₹1,92,000	Painting & Coatings	₹99,840
Roofing Works	₹3,12,000	Finishing Works	₹3,75,240
Government Approvals	₹1,50,000	Miscellaneous Expenses	₹1,78,066
Contingency Amount (5%)	₹3,22,739	GST (18% applicable)	₹12,19,954
Grand Total Estimate	₹79,97,479		

Construction Timeline & Project Schedule

Construction Phase / Stage	Estimated Duration (Weeks)
Site Preparation	2
Foundation	4
Column Work	3
Wall Construction	4
Roofing	3
Electrical Work	2
Plumbing	2
Flooring	3
Painting	2
Finishing	3
Overall Estimated Duration	7 Months

AI Construction Risk Scorecard

Risk category	Level	Telemetry Diagnosis	Mitigation Recommendation
Delay Risk	Low	Slight logistical supply delay risks or approval buffer slip.	Maintain a 14-day schedule buffer.
Budget Risk	Low	Market steel price volatility.	Procure major materials at project kickoff.
Safety Risk	Medium	Scaffolding falls and heavy vehicle movements.	Enforce mandatory hard-hats and harness usage.
Material Sourcing	Medium	Sourcing delays for premium finishes.	Source from multiple supplier agreements.
Weather Risk	Low	Seasonal monsoon rainfall interruptions.	Perform roofing concrete pours before peak rainy season.

AI Sourcing & Value Optimization Recommendations

- Suggested Workforce: 20 active workers (skilled & unskilled)
- Suggested Machinery: Concrete Mixer Machine, Slab Vibrator, JCB Excavator
- Cost Optimization: Optimize cut-lists for steel rebar to minimize waste below 5%.
- Construction Sequence: Run electrical conduiting concurrently with brick laying partitions.
- Material Sourcing Improvements: Use fly-ash blended OPC cement for columns foundation slab to lower footprint.
- Safety Program: Conduct safety toolboxes each morning for working-at-height laborers.

Final Conclusion: This cost and quantity report is generated based on standard mechanical layouts and area bounding boxes detected from the building blueprint. The quantities and rates comply with current Central Public Works Department (CPWD) schedule indices in India and are ready for commercial execution review.

Marcus Aurelius
Project Manager

Sarah Administrator
Auditor Coordinator